

	Flexibility	Reliability	Cost
Biomass supply			
Agriculture & forest residues	Soil carbon loss	Biodiversity loss	Disperse, low density, ununif...
	Competing markets	Nitrogen leaching due to over...	Competing markets
Lignocellulosic crops	Spread of invasive species D...	Direct & indirect land use cha...	Low yield (high production co...
Organic wastes & lipids	Limited bulk availability	Variable quality	Collection and sorting
Conversion pathways			
Fermentation	Co-location with existing infra...	Establishing adequate operat...	Establishing adequate operat...
Gasification		Gas conditioning and clean u...	Uncertain production costs
Pyrolysis	Pyrolysis oil upgrading and/or...	Catalysts with improved sele...	Pyrolysis oil upgrading and/or...
Upgrading and hydrotreatment	Co-location with existing infra...	Establishing adequate operat...	Establishing adequate operat...
End Use			
Aviation	SAF must be drop-in	Competition with biodiesel pr...	International competition with...
	Blending restricted to 50% (s...	Fuel quality critical to ensure ...	Fossil kerosene cheaper than...
Marine	Competition with biodiesel pr...	International competition with...	International competition with...
	Ensuring compatibility of new...	Fuel quality of less refined fu...	Low prices of HFO, MGO
	Dedicated infrastructure need...	High fuel volumes needed for...	Loss of cargo space due to bi...
Road	Use of existing fuel infrastruc...	Access to renewable raw mat...	International competition with...
			Need of compensation for lo...
	Infrastructure and engine mo...	Shortage of sustainable feed...	Price gap between advanced...